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Structural Privacy Vulnerabilities in Quantum Neural Networks

 (/pdf?id=T3eeSNIFUz)

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 Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors
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Abstract:

Quantum neural networks (QNNs) are increasingly considered for near-term learning tasks involving sensitive data, yet the influence of architectural design choices on their privacy risk remains poorly understood. Prior work has shown that QNNs can leak membership information, but it remains unclear where this leakage originates within the QNN pipeline. Is privacy risk primarily a consequence of the trainable ansatz, or can it be shaped upstream by the non-trainable encoder and its induced Hilbert-space geometry? We study this question through QuRiFT (Quantum Risk and Inference Fault-line Tracer), a controlled audit framework for structural privacy analysis in QML. Our central hypothesis is that the classical-to-quantum encoder shapes the Hilbert-space geometry on which the variational ansatz operates, inducing train-test asymmetries that later become exploitable by membership inference attacks. Using QuRiFT, we perform controlled interventions across QNN, HQNN, and QCNN models by varying feature-map family, data re-uploading repetitions, circuit width, variational depth, and variational-layer design while keeping training protocols fixed. The results reveal a clear structural hierarchy in our controlled regimes: privacy risk is driven more strongly by the data-encoding circuit than by trainable variational depth. In particular, z and zz feature maps induce larger generalization gaps and stronger membership signals than effSU2 . Feature-map repetitions and circuit width act as leakage amplifiers, with repetitions correlating strongly with larger train-test gaps ($r = 0.74$) and lower test accuracy ($r = -0.83$), producing a privacy-utility anti-tradeoff. Direct membership inference validates the structural signal: high-risk configurations identified by QuRiFT reach attack AUCs up to 0.870, compared with 0.530 for low-risk baselines. These findings show that the encoder should be treated as a first-order privacy parameter in QML design, with feature-map choice and data re-uploading audited as security-relevant architectural decisions rather than routine expressivity knobs.

Checklist Confirmation:  I confirm that I have included a paper checklist in the paper PDF.

Financial Support:  Najeeb Ullah (/profile?id=~Najeeb_Ullah1)

Responsible Reviewing:  We acknowledge the responsible reviewing obligations as authors.

Primary Area: General machine learning (e.g., core contributions in supervised and unsupervised methods)
Secondary Area: Deep learning advancements (e.g., architectures, optimizers, representation learning)
Contribution Type: Theory: The main contribution is via theoretical analyses and proofs.
Academic Integrity: 👁 I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (<https://neurips.cc/Conferences/2026/MainTrackHandbook> (<https://neurips.cc/Conferences/2026/MainTrackHandbook>)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.
LLM Usage: 👁 Editing (e.g., grammar, spelling, word choice)
Declaration: 👁 I confirm that the above information is accurate.
Reviewer Nomination: 👁 Muhammad Naveed Aman (/profile?id=~Muhammad_Naveed_Aman1)
Submission Number: 24396

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Meta Review of Submission24396 by Area Chair zyrS

Meta Review by Area Chair zyrS 📅 21 Jul 2026, 19:13 (modified: 24 Jul 2026, 02:16)
👁 Senior Area Chairs, Area Chairs, Authors, Reviewers Submitted, Program Chairs, Area Chair zyrS
📄 Revisions (/revisions?id=aF0m7l8e4X)

Metareview:

The paper studies structural sources of membership leakage in quantum NNs and introduces an approach to examine how encoder and circuit choices affect privacy risk. Reviewers found the encoder-centric perspective interesting and appreciated the controlled empirical analysis. A major limitation, however, is that all experiments are conducted under noiseless simulation, whereas current quantum hardware is inherently noisy and noise may significantly alter, or even mitigate, the observed privacy leakage. It is therefore unclear whether the reported privacy hierarchy would persist under realistic noise conditions.

Reviewers also raised fundamental questions about the experiments and conclusions, for example whether encoder choices independently cause membership leakage or merely induce ordinary overfitting, whether the generalization gap is an adequate proxy for membership risk; whether the selected configurations and limited attack suite are representative; and whether the reported trends are statistically robust and generalize beyond the simple datasets and confounded architecture comparisons considered.

Also, the paper would benefit from a more thorough discussion of privacy in quantum ML and a clearer presentation of recent related work, which would better contextualize the contribution and improve the exposition.

All these concerns limit the contributions and appear to require major revision.

Add: **Author AC Confidential Comment**



Official Review of Submission24396 by Reviewer Epmi

Official Review by Reviewer Epmi 📅 26 Jun 2026, 19:58 (modified: 24 Jul 2026, 00:02)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Epmi

📄 Revisions (/revisions?id=RgUIjAtDd8)

Summary:

This paper studies structural sources of membership leakage in quantum neural networks. The main hypothesis is that the non-trainable classical-to-quantum encoder shapes the Hilbert-space representation before optimization, and that this encoder-induced structure can create train-test asymmetry that later becomes exploitable by membership inference methods. To study this, the paper introduces QuRiFT, a controlled profiling framework that sweeps feature-map family, feature-map repetitions, circuit width, variational depth, entanglement topology, and gate choices across QNN, HQNN, and QCNN models. The paper reports that Z/ZZ feature maps and larger feature-map repetitions tend to produce larger generalization gaps than efficient-SU(2)-style encoders, and that selected high-risk configurations can lead to higher MIA AUC.

The topic is interesting, and the paper raises a useful question: whether QML privacy risk should be treated as a structural design issue rather than only as a consequence of training dynamics. However, I am not fully convinced that the current evaluation isolates the proposed structural source of leakage from generic training-induced leakage due to overfitting.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths

1. Interesting problem framing. The paper studies membership leakage from a QML architecture-design perspective. Instead of only asking whether QNNs leak membership, it asks which circuit components may induce the leakage. This is a meaningful and potentially useful angle.

2. Useful controlled profiling framework. QuRIFT separates feature-map family, feature-map repetitions, circuit width, variational depth, and lower-level variational-layer choices under a fixed training protocol. The distinction between repeated data encoding and variational depth is especially useful, because repetitions inject input-dependent structure through the fixed encoder, whereas variational depth mainly changes trainable capacity.

Weaknesses

1. The evaluation does not sufficiently isolate the proposed structural source of leakage from generic training-induced leakage. The central claim is that encoder-side structural choices, especially feature-map family and data-reuploading repetitions, are important sources of membership leakage. However, the current evidence mostly shows that some structural configurations have larger train-test gaps, and that selected high-gap configurations can have higher MIA AUC. This is not enough to establish that the MIA signal is caused by the proposed structural factors rather than by ordinary overfitting after training.

This distinction matters because membership inference methods exploit differences in output behavior between training and non-training samples. Since the attack features used in this paper include loss, confidence, margin, entropy, and correctness, a higher MIA AUC may simply reflect a larger train-test gap or poorer calibration. In other words, the current results show that some trained models leak more, but they do not clearly show that the encoder/repetition choices independently cause the leakage.

A stronger evaluation would include pairwise comparisons where all other factors are fixed or matched, and only the proposed structural variable changes. For example, compare two models with the same architecture, same width, same variational depth, same training protocol, similar train accuracy, similar test accuracy, and similar generalization gap, but different feature-map family or repetition count. If the MIA AUC still differs, then the claim that these structural choices independently affect membership leakage would be much stronger. Conversely, if the difference disappears after matching the gap, then the contribution should be reframed as identifying structural factors that induce overfitting, not as directly validating structure-induced membership leakage.

2. The paper relies heavily on the generalization gap as a privacy-risk proxy, but generalization gap is not equivalent to membership leakage. The train-test accuracy gap is a useful first-order indicator of overfitting, but it is a dataset-level aggregate metric. MIA is a sample-level inference problem and depends on the separability of member and non-member output distributions, confidence tails, calibration, class-specific behavior, and the specific attack features. Therefore, a large gap does not necessarily imply strong MIA, and a small gap does not rule out membership leakage on rare samples, hard examples, or specific classes.

The paper itself reports exceptions where the gap is not a deterministic predictor of MIA success. This weakens the interpretation of the selected stress/hard regimes. The authors should analyze the relationship between MIA AUC and generalization gap more directly. For example, they could show MIA AUC versus generalization gap across a broader set of configurations, mark feature-map family and repetition count, and explicitly check whether configurations with similar gaps can have different MIA AUCs, or whether some low-gap configurations still show non-trivial MIA. This would clarify whether feature-map choice and repetitions affect membership leakage beyond simply increasing the train-test gap.

3. The MIA evaluation is limited to one access setting and one attack-feature family. The paper evaluates what the authors describe as a black-box MIA, but this setting still assumes access to the full prediction vector. The attack

uses features derived from confidence, entropy, margin, loss, and correctness when the label is known. This is a valid score/probability-vector access setting, but it does not cover the broader range of MIA methods that may emphasize different leakage signals.

In particular, the paper does not separately report simple threshold methods based on loss, confidence, entropy, or margin. This makes it hard to tell which output signal is actually responsible for the reported MIA success. It also does not evaluate calibrated methods using reference or shadow models, which can behave differently from raw score-based attacks, or label-only methods, which test a weaker API setting where probabilities are unavailable.

This matters because different MIA methods measure different properties of the model. Loss-based methods are closely tied to train/test loss separation; confidence and entropy methods are affected by calibration and posterior sharpness; reference-model methods compare the target output against in/out distributions; and label-only methods depend more on decision-boundary robustness. Therefore, the current results do not show whether the reported structural privacy trends are robust across attack settings, or whether they are specific to the chosen full prediction-vector feature set.

I recommend adding separate threshold results for loss, confidence, entropy, and margin, and adding at least one calibrated reference/shadow-model method or one label-only method if feasible. Reporting both AUC and low-FPR TPR for these methods would clarify which membership signals are actually affected by feature-map choice and repetitions.

4. The selected MIA regimes and main trends need stronger statistical support. The baseline/stress/hard regimes are useful for illustration, but they are selected after observing structural overfitting patterns. This makes it difficult to assess whether the observed attack AUC is representative across the full design space. A more convincing approach would run the same MIA suite over all configurations, followed by a regression analysis of attack AUC. The regression should include feature-map family, repetitions, width, depth, train accuracy, test accuracy, generalization gap, gate count, and number of trainable parameters. In addition, the paper should report uncertainty for the main plots, attack AUCs, correlations, and factor-attribution results. Error bars, confidence intervals, or multi-seed/multi-split results are important here because the reported trends may be sensitive to train/test splits, initialization, attack-model training, and the selected configurations.
5. The central Hilbert-space geometry mechanism remains indirect. The paper argues that the encoder induces a Hilbert-space geometry that leads to train-test asymmetry and membership leakage. However, the experiments mainly measure downstream quantities such as generalization gap, attack AUC, and factor correlations. These are useful indicators but do not directly show how the induced kernel geometry changes. The paper would be stronger with direct measurements of train-train, train-test, and test-test Hilbert-Schmidt similarity distributions, kernel alignment with labels, nearest-neighbor concentration, kernel spectrum/effective rank, or margin distributions under different feature maps and repetition counts.
6. The architecture-level and simulation settings limit the scope of the conclusions. The QNN, HQNN, and QCNN settings differ not only in quantum architecture but also in preprocessing and classical heads. Dense QNN uses non-trainable compression, HQNN uses a trainable CNN bottleneck, and QCNN uses a quantum-filter front-end plus an MLP head. Therefore, the observed dampening effect may come from the classical preprocessor, the head capacity, or the latent representation, not only from quantum architectural inductive bias. The paper should report parameter counts and add matched-parameter or frozen-preprocessor ablations.

I also agree that noiseless simulation is useful for isolating encoder effects. However, repetitions and ZZ-style encoders change circuit depth and gate count, and hardware noise or finite-shot effects could change both utility and privacy behavior. The paper should either limit its conclusions to noiseless controlled regimes or include a small finite-shot/noise-model sanity check.

Quality: 3: good

Clarity: 2: not good

Significance: 3: good

Originality: 3: good

Questions:

1. Can the authors provide a clean pairwise ablation where all settings are fixed or matched, and only the feature-map family or repetition count changes before measuring MIA AUC?
2. Can the authors analyze the relationship between generalization gap and MIA more directly? In particular, are there configurations with similar train/test behavior or similar generalization gap but noticeably different MIA AUC, or configurations with small generalization gap but non-trivial MIA risk? Such an analysis would clarify whether the proposed structural factors affect membership leakage beyond simply increasing the train-test gap.
3. Can the authors clarify how representative the selected baseline/stress/hard regimes are within the full sweep? Additional MIA results on a broader subset of configurations, together with uncertainty estimates for the main attack AUCs, would help assess whether the reported trends are stable.
4. Which specific attack feature drives the reported MIA success? Is it loss, confidence, entropy, margin, correctness, or their combination? Please report separate threshold methods for these signals and compare with calibrated reference-model, shadow-model, or label-only methods if feasible.
5. Can the authors directly measure the induced Hilbert-space geometry, for example through kernel similarity distributions, kernel alignment, nearest-neighbor concentration, or kernel spectrum, rather than relying mainly on the generalization gap?

Limitations:

The authors acknowledge several limitations, including the controlled noiseless simulation setting and the need for broader attacks and direct geometry measurements. This is useful, but the current limitations section does not fully address the main evaluation concern: the experiments do not yet isolate whether the proposed structural factors independently cause membership leakage, as opposed to merely inducing ordinary overfitting that is then exploited by an output-based MIA.

The limitations section should explicitly discuss this causal-attribution issue. It should also clarify that the current black-box setting assumes access to score/probability vectors, and therefore should not be interpreted as covering label-only or query-limited black-box settings. The current conclusions may also not transfer directly to calibrated/reference-model attacks or noisy quantum hardware.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

None

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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Author AC Confidential Comment

Rebuttal



Official Review of Submission24396 by Reviewer 1myw

Official Review by Reviewer 1myw 📅 26 Jun 2026, 13:18 (modified: 24 Jul 2026, 00:02)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 1myw

📄 Revisions (/revisions?id=z7omMThoj2)

Summary:

The paper mainly argues that the non-trainable data encoders in QNNs expose a vulnerable attack surface for privacy leakage, such as membership inference attacks (MIAs). Starting from the abstract itself, the authors make it explicitly clear that certain encoding methods, like Z and ZZ feature maps, along with repeated data re-uploading, increased the vulnerability to MIAs. This vulnerability is claimed to be induced by train-test asymmetries, which are methodically tested via the termed "QuRiFT" framework. So far, the paper presents the experimental validations showing the leakages.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths:

1. The core idea of studying privacy leakages in the very first encoding stage of a QNN, rather than the VQC training part (some related works have done it), is a novel direction.
2. The experimental setup is good with a lot of ablations around encoders, width, entangling ways and gate types. The experiments are controlled and the overall paper is easy to follow.
3. The authors provide empirical evidence of a privacy-utility anti-tradeoff, which was particularly interesting. Also, from the negative-control analysis, it is good to see that the lower-level layers have a lesser effect than the encoding layers.
4. The validation experiments with black-box membership inference attacks with prediction-vector features are coherent with the structural vulnerability claims. For a preliminary qualitative validation, the experiments are largely complete.

Weaknesses:

1. Major: Noiseless Simulation The authors acknowledge this as a limitation of their work and tend to justify the choice of noiseless simulation as a method to get isolated results in their claim. However, opting noise out in this scenario raises concerns, particularly when data leakage is involved. Although I agree that an ideal simulation to

understand the fundamental structural vulnerability is essential, without realistic experiments including noise, the contribution is weak and partial, more like an ablation with initial findings. Interestingly, quantum noise channels have been proposed multiple times to provide different privacy mechanisms and adversarial robustness in QNNs. The severity and even dynamics of vulnerability might be different under these scenarios.

2. A limitation as well as a weakness is the statement that the structural privacy hierarchy identified may not generalize to all possible QML architectures, datasets, or hardware backends.
3. The MIA evaluation seems too narrow for a NeurIPS grade paper. The black box threat model and output-based design are valid for a specific MIA case, but the attack suite still seems limited. Baselines such as threshold attacks, shadow model-based, or maybe LiRA style would strengthen the paper.
4. Datasets considered are too simple.

Quality: 3: good

Clarity: 3: good

Significance: 2: not good

Originality: 3: good

Questions:

Since all experiments are done in noiseless simulation, can the authors show whether the same privacy hierarchy still holds under realistic quantum noise or finite-shot settings?

The paper states that the structural privacy hierarchy may not generalize to all QML architectures, datasets, or hardware backends. Can the authors clarify the exact scope of this hierarchy and which findings they expect to generalize?

Since the datasets are relatively simple, can the authors clarify how far the conclusions can be extended to the sensitive domains used in the motivation, such as healthcare, finance, and public-sector data?

Also, please consider my concerns from the weakness section.

Limitations:

yes.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

An overview figure would help.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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Rebuttal



Official Review of Submission24396 by Reviewer nVBH

Official Review by Reviewer nVBH 📅 25 Jun 2026, 23:57 (modified: 24 Jul 2026, 00:02)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer nVBH

📄 Revisions (/revisions?id=RVsrSApndS)

Summary:

This paper studies structural privacy vulnerabilities in quantum neural networks. The central hypothesis is that the fixed classical-to-quantum encoder can shape the Hilbert-space geometry before optimization, inducing train-test asymmetries that later become exploitable by membership inference attacks. To investigate this, the paper introduces QuRIFT, a controlled auditing framework that varies feature-map family, feature-map repetitions, circuit width, variational depth, entanglement topology, and gate choices across QNN, HQNN, and QCNN models. The experiments are conducted on synthetic binary datasets and a compressed four-class MNIST subset. The paper reports that Z and ZZ feature maps, feature-map repetitions, and circuit width are more strongly associated with generalization gaps and membership-inference leakage than variational depth or lower-level variational-layer choices. The main conclusion is that encoder design and repeated data encoding should be treated as privacy-relevant architectural decisions in QML.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths

1. Important and timely problem. The paper addresses privacy leakage in near-term quantum machine learning models, with a focus on membership inference. This is a relevant problem, especially because QML is often motivated by applications involving sensitive data.
 2. Interesting encoder-centric framing. A useful aspect of the paper is that it does not treat membership leakage only as a consequence of optimization or trainable model capacity. Instead, it studies whether the fixed classical-to-quantum encoder itself can induce privacy-relevant train-test asymmetries. This is a meaningful perspective for QML model design.
 3. Coherent controlled-audit framework. The proposed QuRIFT framework provides a systematic way to vary feature-map family, feature-map repetitions, circuit width, variational depth, entanglement topology, and gate choices under a controlled training setup. This makes the paper more than a single-model membership-inference study.
 4. Suggestive empirical findings. The reported trends are potentially useful: Z and ZZ feature maps appear to produce larger generalization gaps than effSU2; feature-map repetitions correlate more strongly with overfitting and attack metrics than variational depth; and repeated encoding may reduce test accuracy while increasing leakage. These observations could be valuable for QML practitioners if made statistically robust.
 5. Direct MIA validation. The paper does not rely only on generalization gaps. It also runs output-based black-box membership inference attacks, which is important for validating whether the proposed structural proxy corresponds to actual leakage.
- #### Weaknesses
6. The paper lacks multi-seed results, confidence intervals, error bars, and significance tests, making the main attack and trend claims difficult to assess for robustness.
 7. Regime selection appears post hoc and overly tied to generalization gaps, creating circular validation and weakening claims that structural overfitting reliably predicts membership leakage.

8. Claims about feature maps, encoder geometry, and architectures as leakage amplifiers are correlational, with confounders such as capacity, preprocessing, convergence, and accuracy insufficiently controlled.
9. The study lacks classical and QML baselines, transparent factor-attribution methodology, and broader datasets, limiting attribution and generalization to sensitive-domain settings.

Quality: 2: not good

Clarity: 3: good

Significance: 2: not good

Originality: 3: good

Questions:

1. Can the authors provide multi-seed results with confidence intervals or error bars for the main figures and attack metrics? Are the trends for Z/ZZ versus effSU2, repetitions versus depth, and stress versus hard regimes stable across random initialization, data splits, and MIA attacker training? 2. Can the authors report membership-inference results across all swept configurations, rather than only selected baseline/stress/hard regimes? This would help determine whether the generalization gap is a reliable structural proxy or whether the current validation is affected by post-hoc selection. 3. Can the authors clarify how the factor-attribution percentages are computed? Please specify whether the method is based on ANOVA, Sobol-style variance decomposition, regression, random-forest importance, or another method, and provide uncertainty estimates or sensitivity checks. 4. Can the architecture comparison be made more controlled? For example, can the authors compare QNN, HQNN, and QCNN under matched parameter counts, matched preprocessing capacity, or with classical-only versions of the CNN/MLP components? 5. Can the authors add stronger baselines or controls, such as classical MLP/CNN models, classical kernel methods, quantum kernel classifiers, regularized QNNs, early stopping, differential privacy, or calibration-based defenses?

Limitations:

The authors discuss some limitations, such as noiseless simulation and the need for future work on broader attacks and direct geometry measurements. However, the limitations section does not adequately address several central limitations of the current study: insufficient statistical uncertainty, post-hoc regime selection, lack of classical and stronger QML baselines, confounded architecture comparisons, and narrow datasets relative to the sensitive-domain motivation. I recommend expanding the limitations section to explicitly acknowledge these issues and to weaken claims about sensitive-domain QML deployment unless more realistic datasets and baselines are added.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

N/A

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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